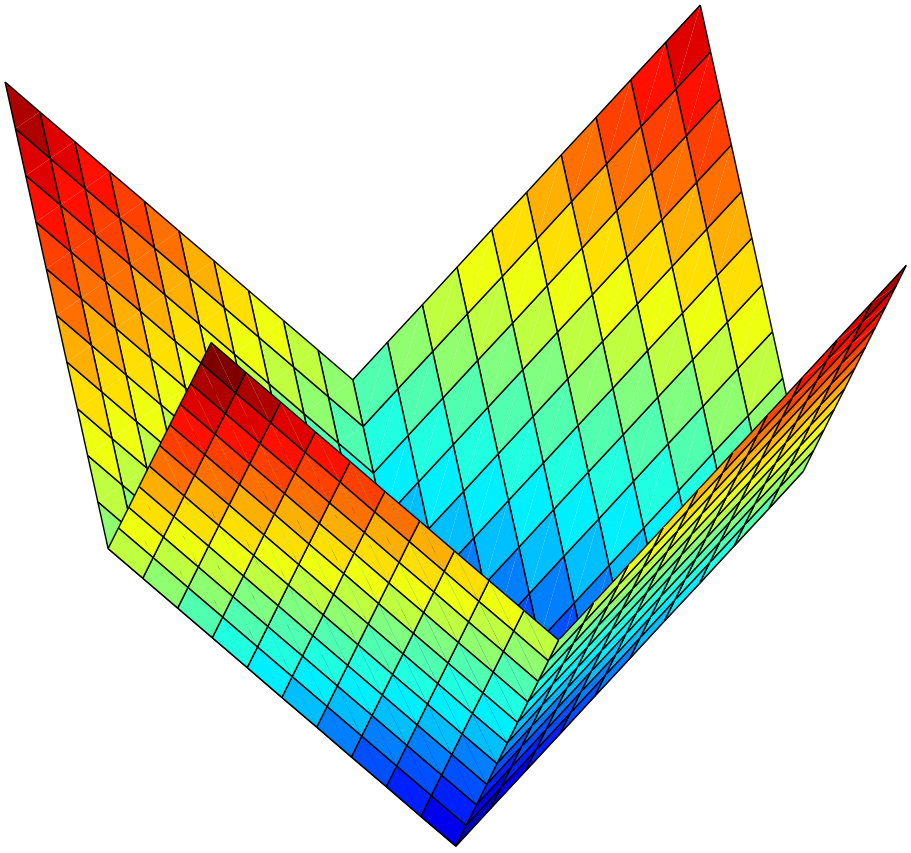
Data Sciences – CentraleSupelec
Large Scale and Distributed Optimization
Part III: Subdifferential and conjugate

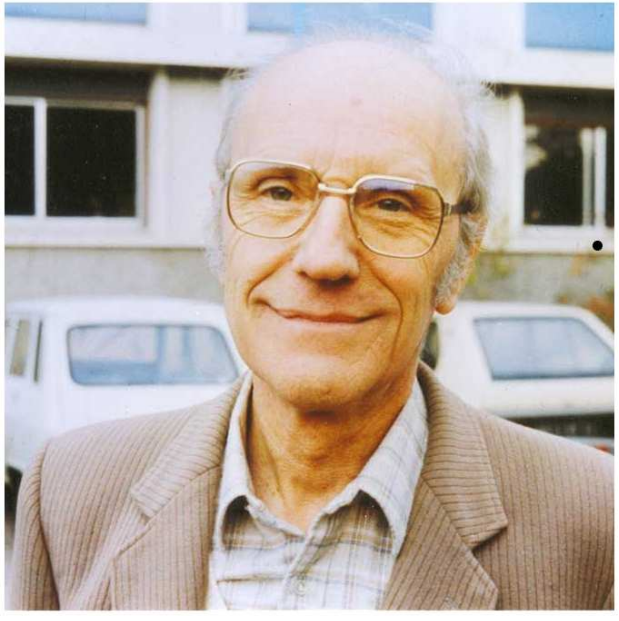
Emilie Chouzenoux and Jean-Christophe Pesquet

Center for Visual Computing
CentraleSupelec
emilie.chouzenoux@u-pem.fr, jean-christophe@pesquet.eu

Non-smooth convex optimization



A pioneer



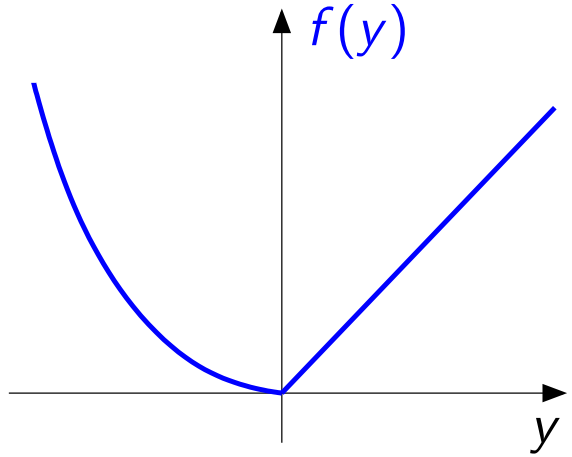
Jean-Jacques Moreau
(1923–2014)

Subdifferential of function: definition

The (Moreau) subdifferential of f, denoted by ∂f,

Subdifferential of function: definition

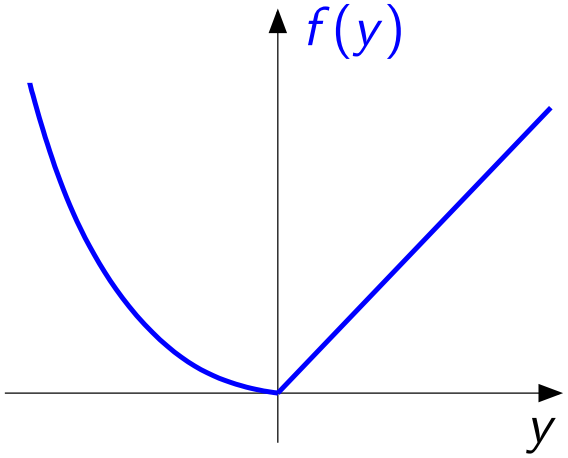
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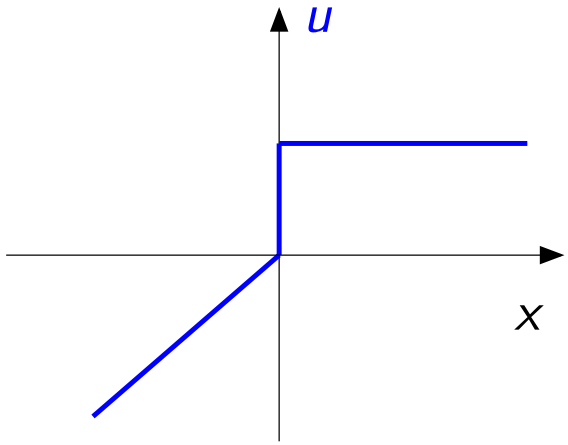


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Let $f : \mathcal{H} \to]-\infty, +\infty]$ be a proper function.
The (Moreau) subdifferential of f, denoted by ∂f, is such that

$$\partial f : \mathcal{H} \to 2^{\mathcal{H}}$$
$$x \to \{u \in \mathcal{H} \,|\, (\forall y \in \mathcal{H})\ \langle y - x | u \rangle + f(x) \leq f(y)\}$$

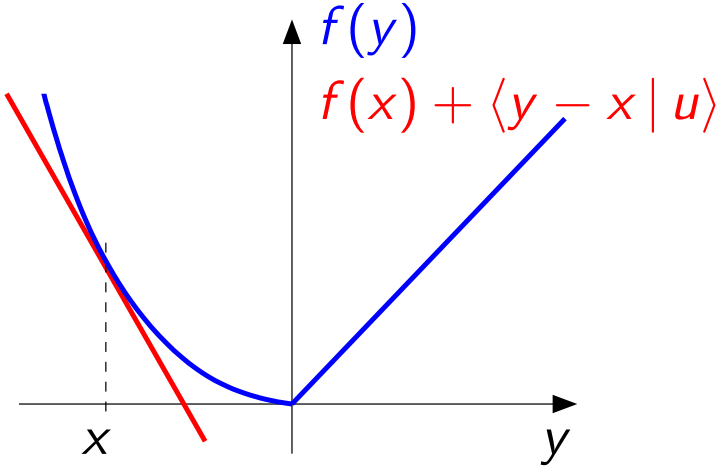


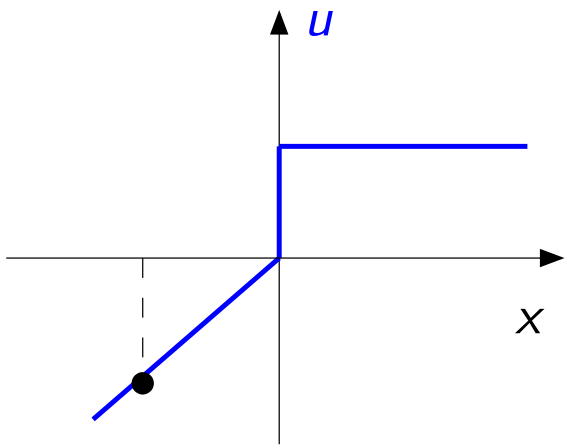


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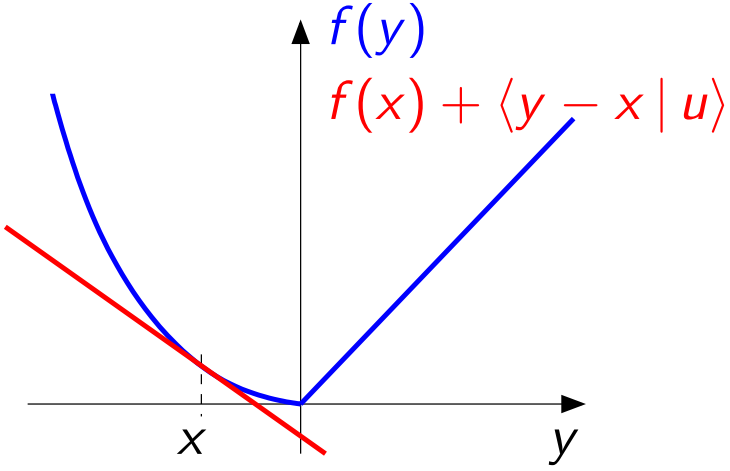


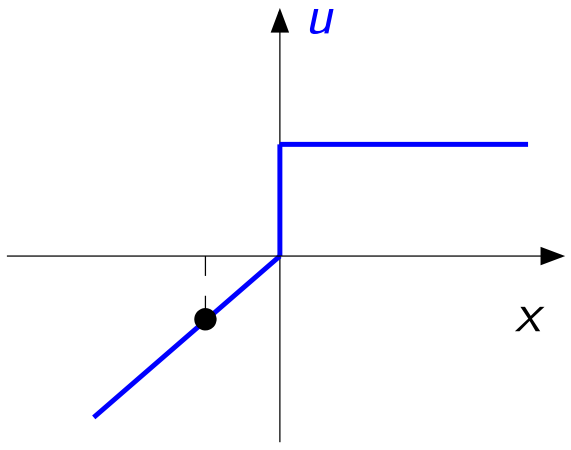


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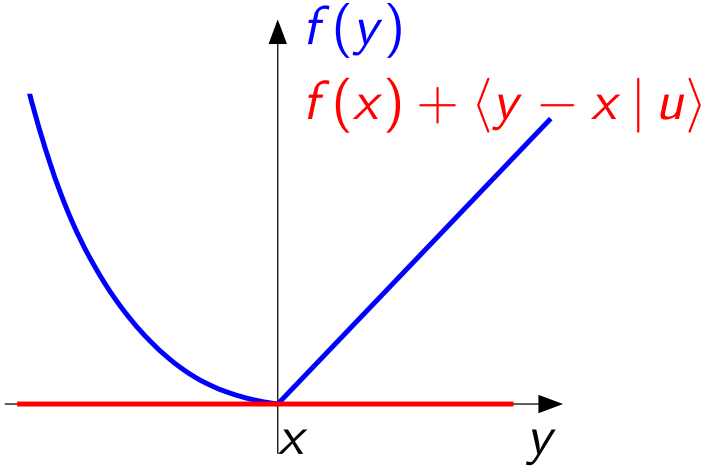


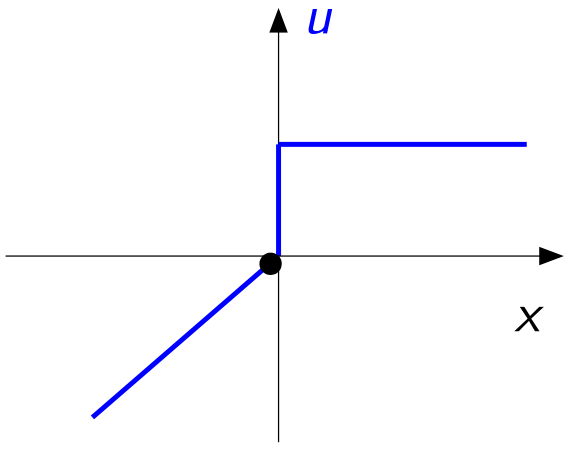


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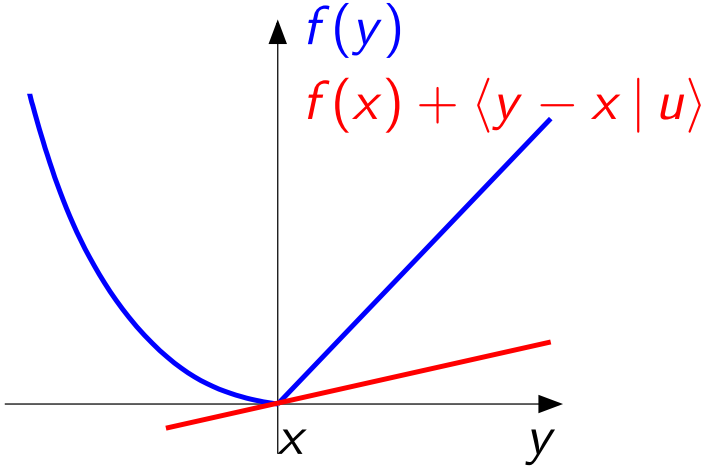


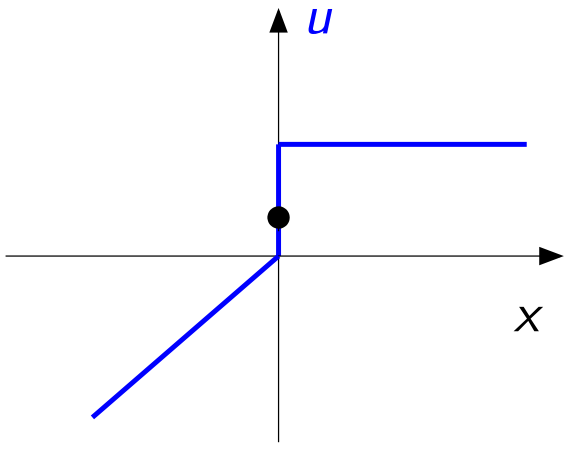


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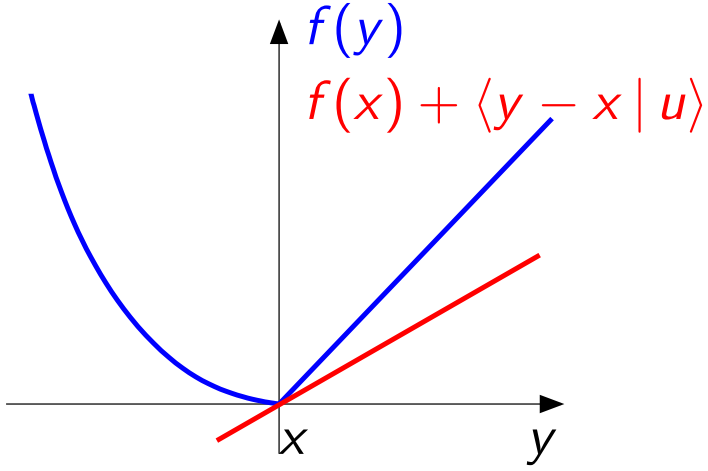


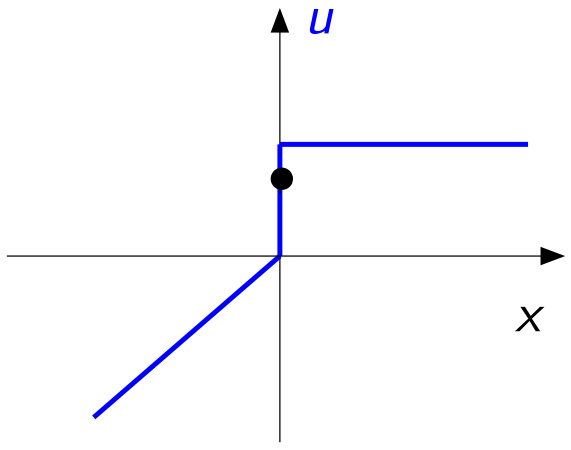


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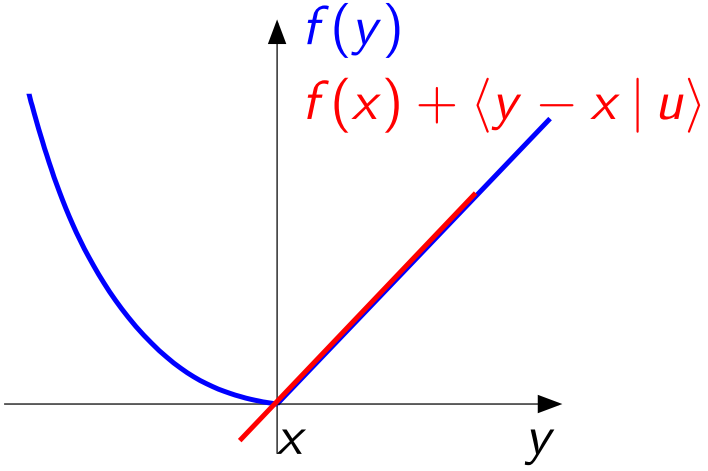


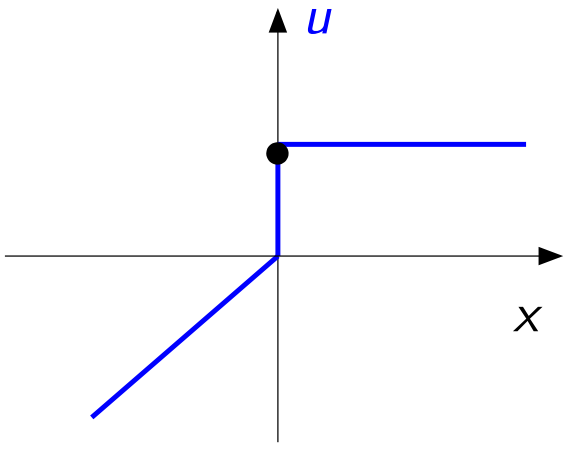


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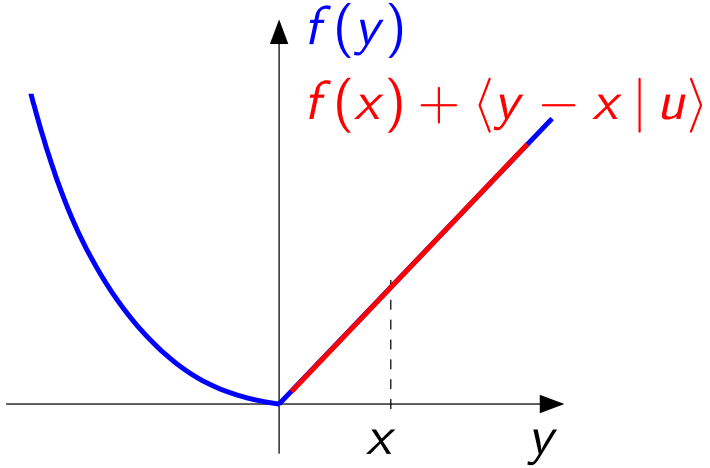


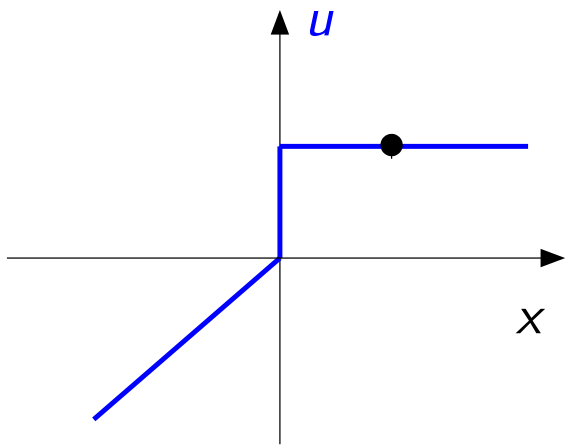


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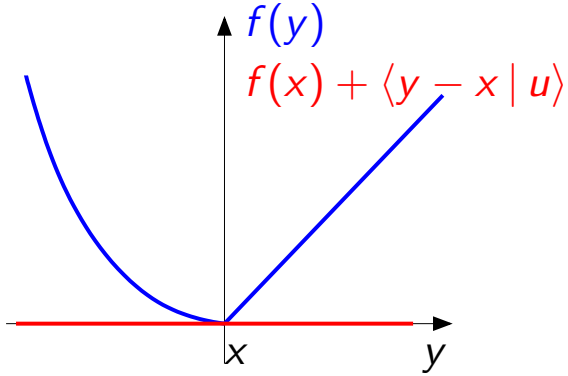


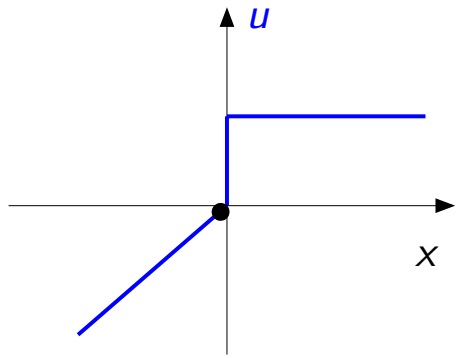


Subdifferential of a function: properties

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Fermat's rule : $0 \in \partial f(x) \Leftrightarrow x \in \operatorname{Argmin} f$

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- For every $x \in \operatorname{dom} f$, $\partial f(x)$ is a closed and convex set.

Subdifferential of a function: properties

Let $f : \mathcal{H} \to]-\infty, +\infty]$ be a proper function.
Its subdifferential is a monotone operator, i.e.

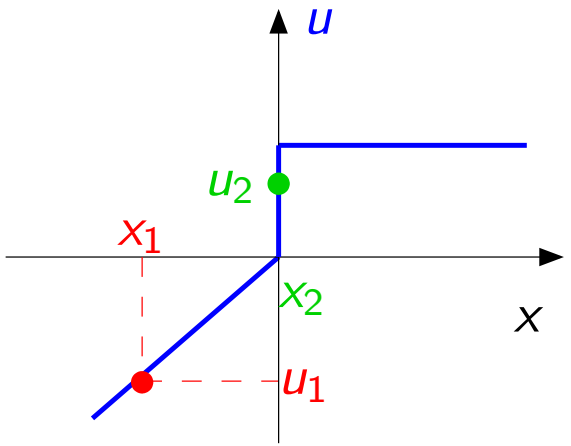
$$\big(\forall (x_1, x_2) \in \mathcal{H}^2\big)\big(\forall u_1 \in \partial f(x_1)\big)\big(\forall u_2 \in \partial f(x_2)\big) \quad \langle u_1 - u_2 \mid x_1 - x_2 \rangle \geq 0.$$

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- Proof:
  By definition:
  $$\langle x_2 - x_1 | u_1 \rangle + f(x_1) \leq f(x_2)$$
  $$\langle x_1 - x_2 | u_2 \rangle + f(x_2) \leq f(x_1)$$
- It results that $\boxed{\langle x_1 - x_2 \,|\, u_1 - u_2 \rangle \geq 0}$.



Subdifferential of a convex function: properties

If $f : \mathcal{H} \to]-\infty, +\infty]$ is convex and it is Gâteaux differentiable at x, then

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$$(\forall y \in \mathcal{H}) \qquad \langle \nabla f(x) \mid y \rangle = \lim_{\substack{\alpha \to 0 \\ \alpha \neq 0}} \frac{f(x + \alpha y) - f(x)}{\alpha}.$$

Proof:
For every $\alpha \in [0, 1]$ and $y \in \mathcal{H}$,

$$f\big(x + \alpha(y - x)\big) \leq (1 - \alpha) f(x) + \alpha f(y)$$

$$\Rightarrow \quad \langle \nabla f(x) \mid y - x \rangle = \lim_{\substack{\alpha \to 0 \\ \alpha > 0}} \frac{f\big(x + \alpha(y - x)\big) - f(x)}{\alpha} \leq f(y) - f(x)$$

Then $\nabla f(x) \in \partial f(x)$.

Subdifferential of a convex function: properties

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Proof:
Conversely, if $u \in \partial f(x)$, then, for every $\alpha \in [0, +\infty[$ and $y \in \mathcal{H}$,

$$f(x+\alpha y) \geq f(x) + \langle u \mid x + \alpha y - x \rangle$$

$$\Rightarrow \quad \langle \nabla f(x) \mid y \rangle = \lim_{\substack{\alpha \to 0 \\ \alpha > 0}} \frac{f(x+\alpha y) - f(x)}{\alpha} \geq \langle u \mid y \rangle$$

By selecting $y = u - \nabla f(x)$, it results that $\|u - \nabla f(x)\|^2 \leq 0$ and then $u = \nabla f(x)$.

Subdifferential of a convex function: properties

Let $f : \mathcal{H} \to]-\infty, +\infty]$ be Gâteaux differentiable on $\operatorname{dom} f$, which is an open convex set.
Then, f is convex if and only if

$$(\forall (x, y) \in (\operatorname{dom} f)^2) \quad f(y) \geq f(x) + \langle \nabla f(x) \mid y - x \rangle .$$

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Proof:
We have already seen that the gradient inequality holds when f is convex and differentiable at $x \in \mathcal{H}$.

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Proof:
Conversely, if the gradient inequality is satisfied, we have, for every $(x, y) \in (\operatorname{dom} f)^2$ and $\alpha \in [0, 1]$, $\alpha x + (1 - \alpha) y \in \operatorname{dom} f$, and

$$f(x) \geq f(\alpha x + (1 - \alpha) y) + (1 - \alpha) \langle \nabla f(\alpha x + (1 - \alpha) y) \mid x - y \rangle$$
$$f(y) \geq f(\alpha x + (1 - \alpha) y) + \alpha \langle \nabla f(\alpha x + (1 - \alpha) y) \mid y - x \rangle .$$

By multiplying the first inequality by α and the second one by $1 - \alpha$ and summing them, we get

$$\alpha f(x) + (1 - \alpha) f(y) \geq f(\alpha x + (1 - \alpha) y).$$

Subdifferential of a convex function: properties

Let $f : \mathcal{H} \to]-\infty, +\infty]$ be Gâteaux differentiable on $\operatorname{dom} f$, which is an open convex set.
Then, f is convex if and only if ∇f is monotone on $\operatorname{dom} f$, i.e.

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Proof:
When f is convex, we have seen that its subdifferential is monotone and, for every $x \in \operatorname{dom} f$, $\partial f(x) = \{\nabla f(x)\}$.

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Proof:
Conversely, assume that ∇f is monotone on $\operatorname{dom} f$. For every $(x, y) \in (\operatorname{dom} f)^2$, let $\varphi \colon [0, 1] \to \mathbb{R} \colon \alpha \mapsto f(x + \alpha(y - x))$.
φ is differentiable on $[0, 1]$ and

$$(\forall \alpha \in [0, 1]) \qquad \varphi'(\alpha) = \langle \nabla f(x + \alpha(y - x)) \mid y - x \rangle .$$

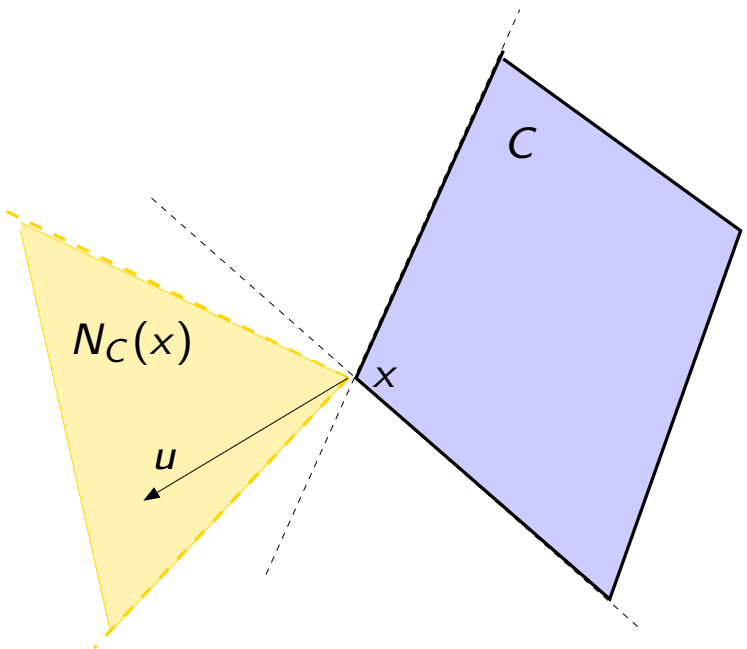
On the other hand, for every $\alpha \in]0, 1]$

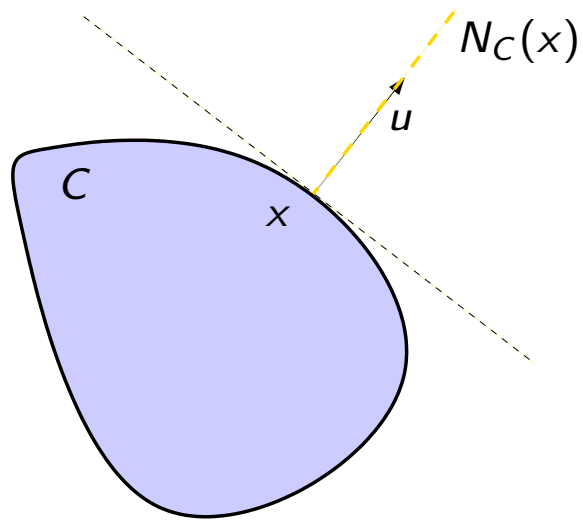
$$\begin{aligned}
& \langle \nabla f(x + \alpha(y - x)) - \nabla f(x) \mid y - x \rangle \geq 0 \\
\Leftrightarrow \quad & \varphi'(\alpha) \geq \langle \nabla f(x) \mid y - x \rangle \\
\Rightarrow \quad & \varphi(1) - \varphi(0) = \int_0^1 \varphi'(\alpha) d\alpha \geq \langle \nabla f(x) \mid y - x \rangle \\
\Leftrightarrow \quad & f(y) - f(x) \geq \langle \nabla f(x) \mid y - x \rangle .
\end{aligned}$$

Subdifferential of a convex function: example

Let C be a nonempty subset of $\mathcal{H}$.
For every $x \in \mathcal{H}$, $\partial \iota_C(x)$ is the normal cone to C at x defined by

$$N_C(x) = \begin{cases} \{u \in \mathcal{H} \mid (\forall y \in C) \ \ \langle u \mid y - x \rangle \leq 0\} & \text{if } x \in C \\ \varnothing & \text{otherwise.} \end{cases}$$





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- If $x \in \operatorname{int} C$, then $N_C(x) = \{0\}$.
- If C is a vector space, then for every $x \in C$, $N_C(x) = C^{\perp}$.
- Let $c \in \mathcal{H}$, $\rho \in \,]0, +\infty[$ and $C = \overline{\mathcal{B}}(c, \rho) = \{y \in \mathcal{H} \mid \|y - c\| \leq \rho\}$.
  For every $x \in C$,

$$N_C(x) = \begin{cases} \{\alpha(x - c) \mid \alpha \in [0, +\infty[\} & \text{if } \|x - c\| = \rho \\ \{0\} & \text{si } \|x - c\| < \rho. \end{cases}$$

Subdifferential calculus

Let $\mathcal{H}$ and $\mathcal{G}$ be two real Hilbert spaces.

- Let $f : \mathcal{H} \to]-\infty, +\infty]$ be proper, then for every $\lambda \in]0, +\infty[$ $\partial(\lambda f) = \lambda \partial f$.
- Let $f : \mathcal{H} \to]-\infty, +\infty]$, $g : \mathcal{G} \to]-\infty, +\infty]$, and $L \in \mathcal{B}(\mathcal{H}, \mathcal{G})$. If $\operatorname{dom} g \cap L(\operatorname{dom} f) \neq \varnothing$, then

$$(\forall x \in \mathcal{H}) \qquad \partial f(x) + L^* \partial g(Lx) \subset \partial(f + g \circ L)(x).$$

Subdifferential calculus

Let $\mathcal{H}$ and $\mathcal{G}$ be two real Hilbert spaces.

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$$(\forall x \in \mathcal{H}) \qquad \partial f(x) + L^* \partial g(Lx) \subset \partial(f + g \circ L)(x).$$

Proof: Let $x \in \mathcal{H}$, $u \in \partial f(x)$ and $v \in \partial g(Lx)$. We have: $u + L^* v \in \partial f(x) + L^* \partial g(Lx)$ and

$$(\forall y \in \mathcal{H}) \qquad \begin{aligned} f(y) &\geq f(x) + \langle y - x \mid u \rangle \\ g(Ly) &\geq g(Lx) + \langle L(y - x) \mid v \rangle . \end{aligned}$$

Therefore, by summing,

$$f(y) + g(Ly) \geq f(x) + g(Lx) + \langle y - x \mid u + L^* v \rangle .$$

We deduce that $u + L^* v \in \partial(f + g \circ L)(x)$.

Subdifferential calculus

Let $\mathcal{H}$ and $\mathcal{G}$ be two real Hilbert spaces.
Let $f \in \Gamma_0(\mathcal{H})$, $g \in \Gamma_0(\mathcal{G})$, and $L \in \mathcal{B}(\mathcal{H}, \mathcal{G})$.
If $\operatorname{int}(\operatorname{dom} g) \cap L(\operatorname{dom} f) \neq \varnothing$ or $\operatorname{dom} g \cap \operatorname{int}\big(L(\operatorname{dom} f)\big) \neq \varnothing$, then

$$\partial f + L^* \partial g \, L = \partial(f + g \circ L) \, .$$

Particular case:

- If $f \in \Gamma_0(\mathcal{H})$, $g \in \Gamma_0(\mathcal{H})$, and g is finite valued, then $\partial f + \partial g = \partial(f + g)$.
- If $g \in \Gamma_0(\mathcal{G})$, $L \in \mathcal{B}(\mathcal{H}, \mathcal{G})$, and $\operatorname{int}(\operatorname{dom} g) \cap \operatorname{ran} L \neq \varnothing$, then $L^* \partial g \, L = \partial(g \circ L)$.

Subdifferential calculus

Let $(\mathcal{H}_i)_{i\in I}$ where $I \subset \mathbb{N}$ be Hilbert spaces and let $\mathcal{H} = \bigoplus_{i\in I} \mathcal{H}_i$.
For every $i \in I$, let $f_i : \mathcal{H}_i \to \left]-\infty, +\infty\right]$ be a proper function. Let

$$f : \mathcal{H} \to \left]-\infty, +\infty\right] : x = (x_i)_{i\in I} \mapsto \sum_{i\in I} f_i(x_i)$$

Then,

$$\big(\forall x = (x_i)_{i\in I} \in \mathcal{H}\big) \qquad \partial f(x) = \bigtimes_{i\in I} \partial f_i(x_i).$$

Subdifferential calculus

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Then,

$$\big(\forall x = (x_i)_{i\in I} \in \mathcal{H}\big) \qquad \partial f(x) = \bigtimes_{i\in I} \partial f_i(x_i).$$

Proof: Let $x = (x_i)_{i\in I} \in \mathcal{H}$. We have

$$\begin{aligned}
& t = (t_i)_{i\in I} \in \bigtimes_{i\in I} \partial f_i(x_i) \\
\Leftrightarrow \quad & (\forall i \in I)(\forall y_i \in \mathcal{H}_i) \;\; f_i(y_i) \geq f_i(x_i) + \langle t_i \mid y_i - x_i \rangle \\
\Rightarrow \quad & \big(\forall y = (y_i)_{i\in I} \in \mathcal{H}\big) \;\; \sum_{i\in I} f_i(y_i) \geq \sum_{i\in I} f_i(x_i) + \sum_{i\in I} \langle t_i \mid y_i - x_i \rangle \\
\Leftrightarrow \quad & \big(\forall y \in \mathcal{H}\big) \;\; f(y) \geq f(x) + \langle t \mid y - x \rangle .
\end{aligned}$$

Subdifferential calculus

Let $(\mathcal{H}_i)_{i\in I}$ where $I \subset \mathbb{N}$ be Hilbert spaces and let $\mathcal{H} = \bigoplus_{i\in I} \mathcal{H}_i$.
For every $i \in I$, let $f_i : \mathcal{H}_i \to]-\infty, +\infty]$ be a proper function. Let

$$f : \mathcal{H} \to]-\infty, +\infty] : x = (x_i)_{i\in I} \mapsto \sum_{i\in I} f_i(x_i)$$

Then,

$$\big(\forall x = (x_i)_{i\in I} \in \mathcal{H}\big) \qquad \partial f(x) = \bigtimes_{i\in I} \partial f_i(x_i).$$

Proof: Conversely,

$$\begin{aligned} & t = (t_i)_{i\in I} \in \partial f(x) \\ \Leftrightarrow \quad & \big(\forall y = (y_i)_{i\in I} \in \mathcal{H}\big) \ \sum_{i\in I} f_i(y_i) \geq \sum_{i\in I} f_i(x_i) + \sum_{i\in I} \langle t_i \mid y_i - x_i \rangle . \end{aligned}$$

Let $j \in I$. By setting $(\forall i \in I \setminus \{j\})\ y_i = x_i \in \operatorname{dom} f_i$, we get

$$(\forall y_j \in \mathcal{H}_j) \ \ f_j(y_j) \geq f_j(x_j) + \langle t_j \mid y_j - x_j \rangle .$$

Exercise 1: Huber function

Let $\rho > 0$ and set

$$f : \mathbb{R} \to \mathbb{R} : \mapsto \begin{cases} \frac{x^2}{2}, & \text{if } |x| \leq \rho \\ \rho|x| - \frac{\rho^2}{2}, & \text{otherwise.} \end{cases}$$

1. What is the domain of f ?
2. Is f differentiable ? twice-differentiable ?
3. Prove that f is convex. Is it strictly convex ?

Exercise 2

Let $\mathcal{H}$ be a Hilbert space. Let $f : \mathcal{H} \to]-\infty, +\infty]$ and let $C \subset \mathcal{H}$ such that $\operatorname{dom} f \cap C \neq \varnothing$.

1. Give a sufficient condition for $x \in \mathcal{H}$ to be a global minimizer of $f + \iota_C$.
2. When $f \in \Gamma_0(\mathcal{H})$, C is a closed convex set, and f is Gâteaux-differentiable on $\mathcal{H}$, give a necessary and sufficient condition for $x \in \mathcal{H}$ to be a minimizer of $f + \iota_C$.

Conjugate



Adrien-Marie Legendre
(1752–1833)



Werner Fenchel
(1905–1988)

Conjugate: definition

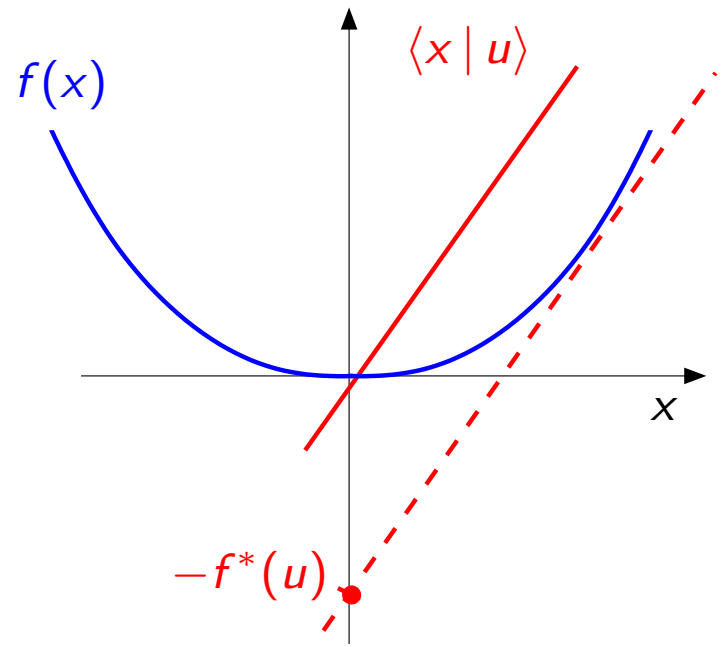
Let $\mathcal{H}$ be a Hilbert space and $f : \mathcal{H} \to]-\infty, +\infty]$.
The conjugate of f is the function $f^* : \mathcal{H} \to [-\infty, +\infty]$ such that

$$(\forall u \in \mathcal{H}) \qquad f^*(u) = \sup_{x \in \mathcal{H}} \big(\langle x \mid u \rangle - f(x) \big) .$$

Conjugate: definition

Let $\mathcal{H}$ be a Hilbert space and $f : \mathcal{H} \to]-\infty, +\infty]$.
The conjugate of f is the function $f^* : \mathcal{H} \to [-\infty, +\infty]$ such that

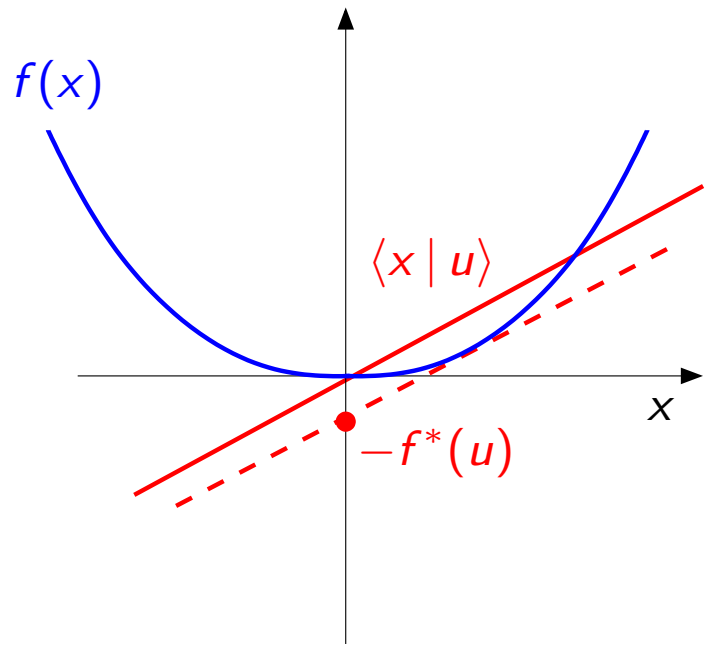
$$(\forall u \in \mathcal{H}) \qquad f^*(u) = \sup_{x \in \operatorname{dom} f} \big(\langle x \mid u \rangle - f(x) \big)$$



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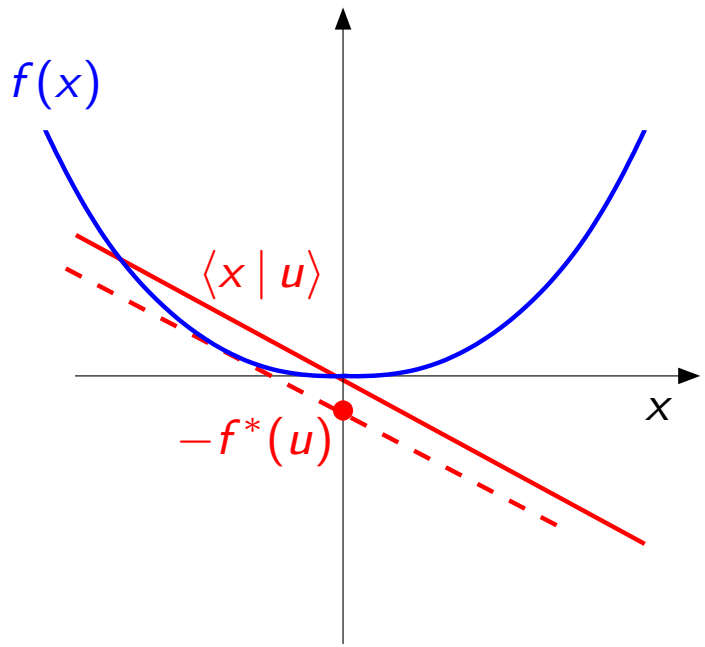
$$(\forall u \in \mathcal{H}) \qquad f^*(u) = \sup_{x \in \operatorname{dom} f} \big(\langle x \mid u \rangle - f(x) \big)$$



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Examples :

- $f = \frac{1}{2}\|\cdot\|^2 \Rightarrow f^* = \frac{1}{2}\|\cdot\|^2$

  Proof : For every $(x, u) \in \mathcal{H}^2$, $\langle x \mid u \rangle - \frac{1}{2}\|x\|^2 = \frac{1}{2}\|u\|^2 - \frac{1}{2}\|u - x\|^2$ is maximum at $x = u$.
  Consequently, $f^*(u) = \frac{1}{2}\|u\|^2$.

Conjugate: definition

Let $\mathcal{H}$ be a Hilbert space and $f : \mathcal{H} \to]-\infty, +\infty]$.
The conjugate of f is the function $f^* : \mathcal{H} \to [-\infty, +\infty]$ such that

$$(\forall u \in \mathcal{H}) \qquad f^*(u) = \sup_{x \in \operatorname{dom} f} \big(\langle x \mid u \rangle - f(x) \big)$$

Examples :

- $f = \frac{1}{2}\|\cdot\|^2 \Rightarrow f^* = \frac{1}{2}\|\cdot\|^2$.
- Let $\phi : \mathbb{R} \to]-\infty, +\infty]$ be a even function. $(\phi \circ \|\cdot\|)^* = \phi^* \circ \|\cdot\|$.
- $(\forall x \in \mathbb{R}^N)\ f(x) = \frac{1}{q}\|x\|_q^q$ with $q \in]1, +\infty[$
  $\Rightarrow (\forall u \in \mathbb{R}^N)\ f^*(u) = \frac{1}{q^*}\|u\|_{q^*}^{q^*}$ with $\frac{1}{q} + \frac{1}{q^*} = 1$

Conjugate: definition

Let $\mathcal{H}$ be a Hilbert space and $f : \mathcal{H} \to]-\infty, +\infty]$.
The conjugate of f is the function $f^* : \mathcal{H} \to [-\infty, +\infty]$ such that

$$(\forall u \in \mathcal{H}) \qquad f^*(u) = \sup_{x \in \operatorname{dom} f} \big(\langle x \mid u \rangle - f(x) \big)$$

- If f is even, then f^* is even.

Proof :

$$\begin{aligned}
(\forall u \in \mathcal{H}) \quad f^*(-u) &= \sup_{x \in \mathcal{H}} \big(\langle x \mid -u \rangle - f(x) \big) \\
&= \sup_{x \in \mathcal{H}} \big(\langle -x \mid -u \rangle - f(-x) \big) \\
&= \sup_{x \in \mathcal{H}} \big(\langle x \mid u \rangle - f(x) \big) \\
&= f^*(u).
\end{aligned}$$

Conjugate: definition

Let $\mathcal{H}$ be a Hilbert space and $f : \mathcal{H} \to]-\infty, +\infty]$.
The conjugate of f is the function $f^* : \mathcal{H} \to [-\infty, +\infty]$ such that

$$(\forall u \in \mathcal{H}) \qquad f^*(u) = \sup_{x \in \operatorname{dom} f} \big(\langle x \mid u \rangle - f(x) \big)$$

- If f is even, then f^* is even.
- For every $\alpha \in]0, +\infty[$, $(\alpha f)^* = \alpha f^*(\cdot/\alpha)$.
- For every $(y, v) \in \mathcal{H}^2$ and $\alpha \in \mathbb{R}$,
  $\big(f(\cdot - y) + \langle \cdot \mid v \rangle + \alpha\big)^* = f^*(\cdot - v) + \langle y \mid \cdot - v \rangle - \alpha$.
- Let $\mathcal{G}$ be a Hilbert space and $L \in \mathcal{B}(\mathcal{G}, \mathcal{H})$ be an isomorphism.
  $(f \circ L)^* = f^* \circ (L^{-1})^*$.
- f^* is l.s.c. and convex.

Conjugate: definition

Let $\mathcal{H}$ be a Hilbert space and $f : \mathcal{H} \to]-\infty, +\infty]$.
The conjugate of f is the function $f^* : \mathcal{H} \to [-\infty, +\infty]$ such that

$$(\forall u \in \mathcal{H}) \qquad f^*(u) = \sup_{x \in \operatorname{dom} f} \big(\langle x \mid u \rangle - f(x) \big)$$

Moreau-Fenchel theorem

Let $\mathcal{H}$ be a Hilbert space and $f : \mathcal{H} \to]-\infty, +\infty]$ be a proper function.

$$f \text{ is l.s.c. and convex} \Leftrightarrow f^{**} = f.$$

Conjugate: definition

Let $\mathcal{H}$ be a Hilbert space and $f : \mathcal{H} \to]-\infty, +\infty]$.
The **conjugate** of f is the function $f^* : \mathcal{H} \to [-\infty, +\infty]$ such that

$$(\forall u \in \mathcal{H}) \qquad f^*(u) = \sup_{x \in \operatorname{dom} f} \big(\langle x \mid u \rangle - f(x) \big)$$

Moreau-Fenchel theorem
Let $\mathcal{H}$ be a Hilbert space and $f : \mathcal{H} \to]-\infty, +\infty]$ be a proper function.

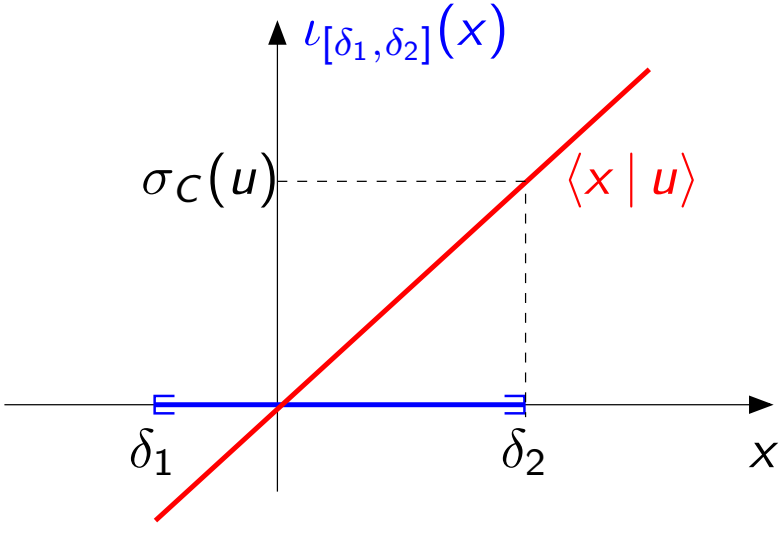
$$f \text{ is l.s.c. and convex} \Leftrightarrow f^{**} = f.$$

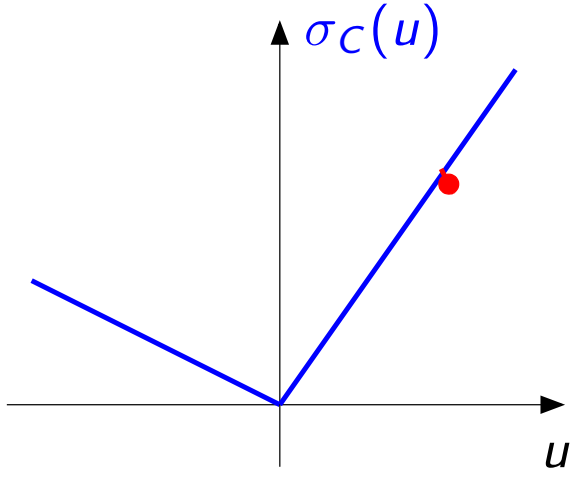
- Consequence: If $f \in \Gamma_0(\mathcal{H})$, then f^* is proper, hence $f^* \in \Gamma_0(\mathcal{H})$.

Conjugate: example

Let $\mathcal{H}$ be a Hilbert space and $C \subset \mathcal{H}$.
σ_C is the support function of C if

$$(\forall u \in \mathcal{H}) \qquad \sigma_C(u) = \sup_{x \in C} \langle x \mid u \rangle$$
$$= \iota_C^*(u).$$

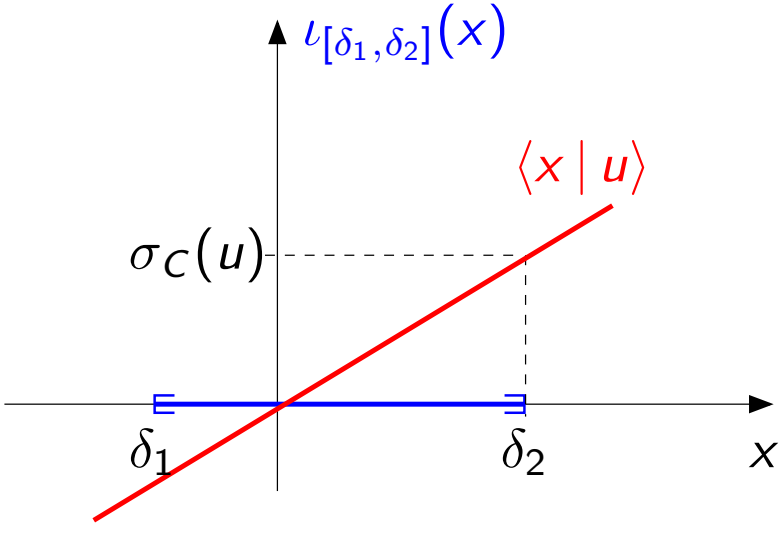


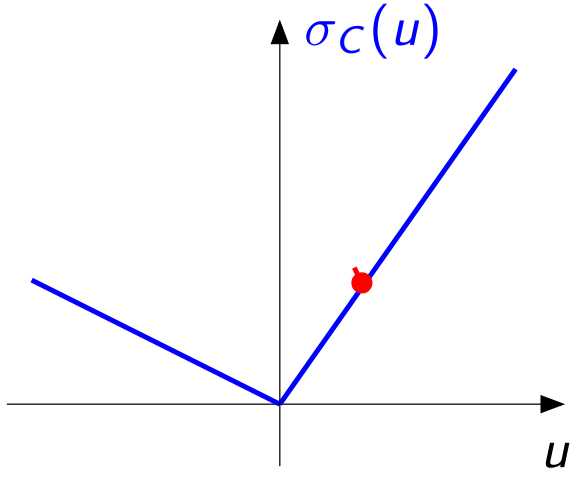


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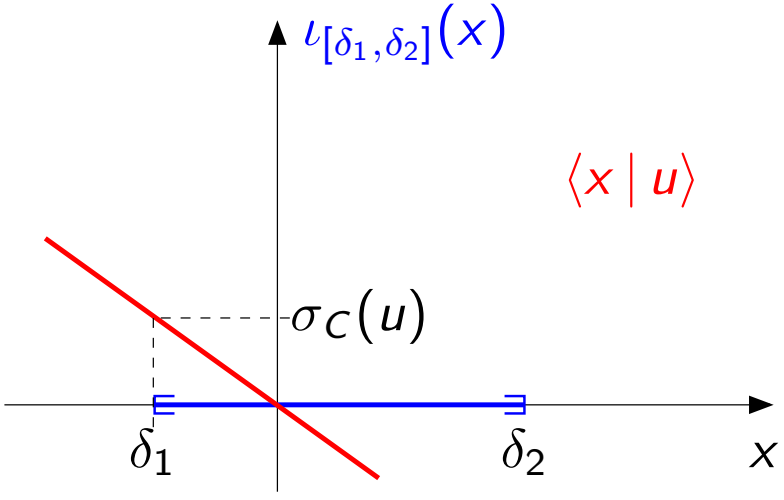


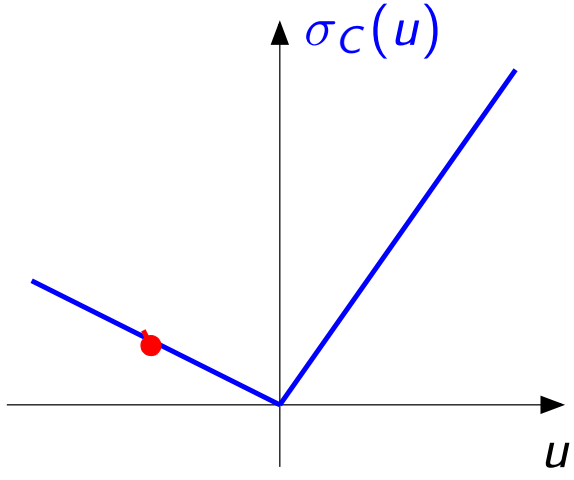


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$$= \iota_C^*(u).$$





Conjugate: example

Let $\mathcal{H}$ be a Hilbert space.
$f : \mathcal{H} \to]-\infty, +\infty]$ is **positively homogeneous** if

$$(\forall x \in \mathcal{H})(\forall \alpha \in]0, +\infty[) \qquad f(\alpha x) = \alpha f(x).$$

f positively homogeneous and belongs to $\Gamma_0(\mathcal{H})$ if and only if $f = \sigma_C$ where C is a nonempty closed convex subset of $\mathcal{H}$.

Conjugate: example

Let $\mathcal{H}$ be a Hilbert space.
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f positively homogeneous and belongs to $\Gamma_0(\mathcal{H})$ if and only if $f = \sigma_C$ where C is a nonempty closed convex subset of $\mathcal{H}$.

Proof: ($\Leftarrow$)
$f = \iota_C^*$ and $\iota_C \in \Gamma_0(\mathcal{H})$. Consequently $\sigma_C \in \Gamma_0(\mathcal{H})$.
Moreover, $(\forall x \in \mathcal{H})\ (\forall \alpha \in]0, +\infty[)\ \sigma_C(\alpha x) = \alpha \sigma_C(x)$.

Conjugate: example

Let $\mathcal{H}$ be a Hilbert space.
$f : \mathcal{H} \to]-\infty, +\infty]$ is positively homogeneous if

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f positively homogeneous and belongs to $\Gamma_0(\mathcal{H})$ if and only if $f = \sigma_C$ where C is a nonempty closed convex subset of $\mathcal{H}$.

Proof : $(\Rightarrow)$
Let $y \in \operatorname{dom} f$.
$f(0) = \lim_{\alpha \to 0, \alpha \geq 0} f\big((1-\alpha)0 + \alpha y\big) = \lim_{\alpha \to 0} \alpha f(y) = 0$.
Let $C = \big\{u \in \mathcal{H} \;\big|\; (\forall x \in \mathcal{H})\, \langle x \mid u \rangle \leq f(x)\big\}$.
We have, for every $u \in C$,
$f^*(u) = \sup_{x \in \mathcal{H}} \langle x \mid u \rangle - f(x) \leq 0 = \langle 0 \mid u \rangle - f(0) \leq f^*(u)$.
Consequently, $f^*(u) = 0$.

Conjugate: example

Let $\mathcal{H}$ be a Hilbert space.
$f : \mathcal{H} \to]-\infty, +\infty]$ is positively homogeneous if

$$(\forall x \in \mathcal{H})(\forall \alpha \in]0, +\infty[) \qquad f(\alpha x) = \alpha f(x).$$

f positively homogeneous and belongs to $\Gamma_0(\mathcal{H})$ if and only if $f = \sigma_C$ where C is a nonempty closed convex subset of $\mathcal{H}$.

Proof: ($\Rightarrow$)
Moreover, for every $u \notin C$, there exists $x \in \mathcal{H}$ such that $\langle x \mid u \rangle > f(x)$. We have then, for every $\alpha \in]0, +\infty[$, $f^*(u) \geq \langle \alpha x \mid u \rangle - f(\alpha x) = \alpha\big(\langle x \mid u \rangle - f(x)\big)$. By taking $\alpha \to +\infty$, we obtain $f^*(u) = +\infty$.
To conclude, $f^* = \iota_C \in \Gamma_0(\mathcal{H}) \Rightarrow f = \sigma_C$ and C is a nonempty closed convex set.

Conjugate: example

- Let $f : \mathbb{R} \to]-\infty, +\infty] : x \mapsto \begin{cases} \delta_1 x & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ \delta_2 x & \text{if } x > 0 \end{cases}$
  with $-\infty \leq \delta_1 < \delta_2 \leq +\infty$.
  Then, $f = \sigma_C$ where C is the closed real interval such that $\inf C = \delta_1$ et $\sup C = \delta_2$.

Conjugate: example

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  Then, $f = \sigma_C$ where C is the closed real interval such that $\inf C = \delta_1$ et $\sup C = \delta_2$.

- Let f be a ℓ^q norm of $\mathbb{R}^N$ with $q \in [1, +\infty]$.
  We have $f = \sigma_C$ where

$$C = \left\{ y \in \mathbb{R}^N \;\middle|\; \|y\|_{q^*} \leq 1 \right\} \qquad \text{with } \frac{1}{q} + \frac{1}{q^*} = 1.$$

Conjugate: example

- Let $f : \mathbb{R} \to]-\infty, +\infty] : x \mapsto \begin{cases} \delta_1 x & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ \delta_2 x & \text{if } x > 0 \end{cases}$
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$$C = \left\{ y \in \mathbb{R}^N \;\middle|\; \|y\|_{q^*} \leq 1 \right\} \qquad \text{with } \frac{1}{q} + \frac{1}{q^*} = 1.$$

Particular case: ℓ^1 norm of $\mathbb{R}^N$: $C = [-1, 1]^N$.

Conjugate: properties

Fenchel-Young inequality: If f is proper, then

1. $\big(\forall (x,u) \in \mathcal{H}^2\big) \qquad f(x) + f^*(u) \geq \langle x \mid u \rangle$

2. $\big(\forall (x,u) \in \mathcal{H}^2\big) \qquad u \in \partial f(x) \Leftrightarrow \quad f(x) + f^*(u) = \langle x \mid u \rangle.$

If $f \in \Gamma_0(\mathcal{H})$, then

$$\big(\forall (x,u) \in \mathcal{H}^2\big) \qquad u \in \partial f(x) \;\Leftrightarrow\; x \in \partial f^*(u)\,.$$

Conjugate: properties

Let $(\mathcal{H}_i)_{i\in I}$ where $I \subset \mathbb{N}$ be Hilbert spaces and let $\mathcal{H} = \bigoplus_{i\in I} \mathcal{H}_i$.
For every $i \in I$, let $f_i : \mathcal{H}_i \to]-\infty, +\infty]$. Let

$$f : \mathcal{H} \to]-\infty, +\infty] : x = (x_i)_{i\in I} \mapsto \sum_{i\in I} f_i(x_i)$$

Then,

$$\big(\forall u = (u_i)_{i\in I} \in \mathcal{H}\big) \qquad f^*(u) = \sum_{i\in I} f_i^*(u_i)\,.$$

Conjugate: properties

Let $(\mathcal{H}_i)_{i\in I}$ where $I \subset \mathbb{N}$ be Hilbert spaces and let $\mathcal{H} = \bigoplus_{i\in I} \mathcal{H}_i$.
For every $i \in I$, let $f_i : \mathcal{H}_i \to]-\infty, +\infty]$. Let

$$f : \mathcal{H} \to]-\infty, +\infty] : x = (x_i)_{i\in I} \mapsto \sum_{i\in I} f_i(x_i)$$

Then,

$$(\forall u = (u_i)_{i\in I} \in \mathcal{H}) \qquad f^*(u) = \sum_{i\in I} f_i^*(u_i)\,.$$

Proof: Let $u = (u_i)_{i\in I} \in \mathcal{H}$. We have

$$\begin{aligned} f^*(u) &= \sup_{x\in\mathcal{H}} \langle x \mid u\rangle - f(x) \\ &= \sup_{x=(x_i)_{i\in I}\in\mathcal{H}} \sum_{i\in I} \langle x_i \mid u_i\rangle - f_i(x_i) \\ &= \sum_{i\in I} \sup_{x_i\in\mathcal{H}_i} \langle x_i \mid u_i\rangle - f_i(x_i) \\ &= \sum_{i\in I} f_i^*(u_i). \end{aligned}$$

Exercise

- For every $c \in \mathbb{R}^N$, give the expression of the conjugate of

$$f : \mathbb{R}^N \to]-\infty, +\infty]$$
$$x \to \langle c \mid x \rangle .$$

- Same question for

$$g : \mathbb{R}^N \to]-\infty, +\infty]$$
$$x \to \iota_{[0,+\infty[^N}(x - c).$$